

The Linearity and Sensitivity of Lateral Effect Position Sensitive Devices— An Improved Geometry

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Abstract—Lateral effect position sensitive devices (PSD's) are gaining in popularity because they are relatively inexpensive yet offer great position resolution over a large sensitive area. In this paper we compare the linearity, sensitivity, and resolution of various lateral effect PSD's by computing as a function of vertical position the output voltage produced by a light spot which tracks a horizontal line. We describe a new lateral effect PSD we refer to as the clover. Results show that the order of descending linearity and resolution is given by duolateral, clover, pincushion, and then tetralateral PSD. The order of descending sensitivity is given by tetralateral, duolateral, clover, and then pincushion PSD.

I. INTRODUCTION

WHEN a p-n junction is illuminated by a light spot, a photopotential arises along each plane of the junction. The photopotential induces photocurrents which distribute toward electrodes on the planes. This phenomenon is called the lateral photoeffect and was first discovered by Schottky [1] in 1930 although it drew little attention until it was rediscovered by Wallmark [2] in 1959. Wallmark designed a device composed of a Ge-In p^+-n junction in which a light beam projected onto the surface causes a position dependent photopotential difference between the contacts [3].

The lateral photoeffect has been investigated by various researchers with device geometry, material composition, and fabrication receiving most of the attention. The geometries most commonly studied include the tetralateral position sensitive device (PSD) shown in Fig. 1, and the duolateral duoaxis PSD shown in Fig. 2. In Figs. 1 and 2 and all subsequent figures of PSD devices the shaded area is the active area of the device, the hatched area the effective (useful) device area, and the dark areas the electrodes. Both the tetralateral and duolateral geometries offer the Cartesian two-dimensional location of a light spot. Recently, novel geometries suitable for other coordinate systems have been discussed, including work by Xing and Boeder [4] on a PSD for measurement of angular position. Lubke *et al.* [5] have suggested a GaAs alternative to sil-

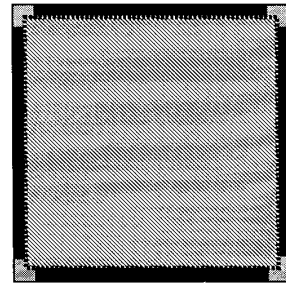


Fig. 1. Tetralateral position sensitive device. All four electrodes are on a single surface. Shaded area is photosensitive, and the heavy solid lines indicate electrodes. The effective device area is shown hatched.

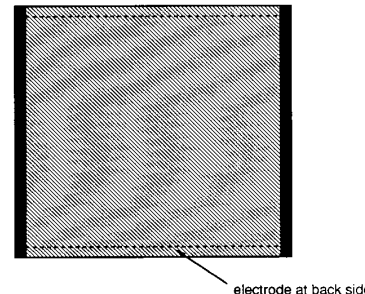


Fig. 2. Duolateral duoaxis position sensitive device. Two of the electrodes are on the p-surface and two on the n-surface.

icon and Ioannou *et al.* [6] have examined gold/beta-silicon carbide Schottky barriers. The fabrication issues for Wallmark-type and duolateral PSD's were addressed by Noorlag and Middelhoek [7]. Subsequently, Schmidt and Ross [8] extended this work to include other lateral effect PSD geometries, and Erb [9] described the associated signal processing circuitry in detail.

The major difference between contemporary lateral effect PSD's and the Wallmark PSD is that modern devices have chosen to detect photocurrents by reverse biasing the junction to saturation in order to prevent forward biasing upon illumination (i.e., by fully reverse-biasing the device) rather than detecting the potential difference. Full reverse-biasing of lateral effect PSD's increases the thickness of the depletion layer between the p and n layers, reducing the current through the device to almost zero,

Manuscript received September 20, 1988; revised April 20, 1989. This work was supported by the Bosque Foundation, by the National Science Foundation, and by the Semiconductor Research Corporation. The review of this paper was arranged by Associate Editor S. D. Senturia.

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IEEE Log Number 8930320.

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and preventing surface recombination. Locovsky [10] derived the fundamental equations describing the diffusion and recombination processes for the steady-state and the small-signal transient cases for modern PSD's. Connors [11] investigated the one-dimensional model of a reverse-biased lateral photodetector. Woltring [12] extended Connors' analysis to the two-dimensional rectangular p-n junction operated in both the unbiased (small signal) and the fully reverse-biased modes. It was shown that fully reverse-biasing results in a much higher resolution, where resolution is defined as the minimum detectable displacement of the light spot on the sensitive surface; higher resolution indicates a smaller minimum detectable displacement.

In his paper, Woltring [12] presented a theoretical analysis of the Wallmark-type, tetralateral, and duolateral duoaxis PSD's. He showed that in the fully reverse-biased mode, the duoaxis duolateral PSD is inherently linear because the x and y contacts are on opposite sides of the junction. Compared to the duolateral PSD, Woltring showed that the tetralateral PSD with four extended ohmic contacts on one side of the p-n junction has disadvantages, namely, the electrode structure makes interference and thus nonlinearity inevitable, and the generated photocurrent is divided into four parts instead of two parts, so that the resolution is roughly half that of the duolateral PSD for the same noise level. However, the tetralateral PSD also has advantages when compared to the duolateral PSD: it has a faster response, a much lower dark current, an easier bias application, and a lower fabrication cost.

An attempt has been made to combine the best aspects of the duolateral PSD with those of the tetralateral PSD by putting all of the contacts on one side of the p-n junction and minimizing the interference between the x and y outputs through careful selection of the boundary and electrode shapes. The result is a commercial device, marketed by Hamamatsu [13] as a pincushion PSD. In this lateral effect PSD, the boundaries are curved and the effective transducer area is formed by a rectangle whose sides are tangent to the innermost points on the edges. The electrical contacts are point contacts located at the four corners of the curved boundaries. This geometry is shown in Fig. 3.

The motivation of the work presented in this paper is to compare the linearity, sensitivity, and resolution of a new type of PSD, referred to as the clover, to the common types of Cartesian PSD's. Noorlag [14] has conducted an analysis of the nonlinearities due to surface inhomogeneities and finite resistance in the electrodes of duolateral PSD's. He concluded that both effects were negligible in most cases, and we assume this to be the case in the analysis presented here. As will be shown, the linearity and resolution of the clover design are better than that of the pincushion and tetralateral PSD's, and the sensitivity of the clover is superior to that of the pincushion but less than that of the tetralateral and duolateral PSD's.

We note that our analysis is not limited to silicon devices, because in the fully reverse-biased mode, the re-

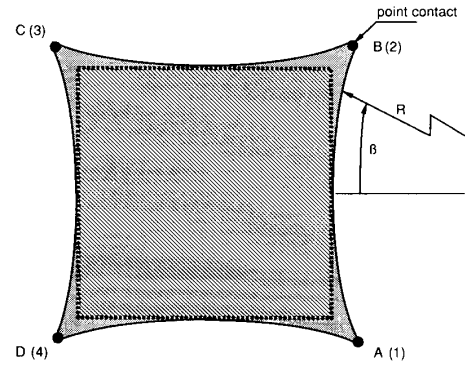


Fig. 3. Pincushion position sensitive device. Boundary curvature is R , focal distance is a , and angular displacement is β .

sponse does not depend on the resistivity of the material as long as it is uniform. The absolute level of the photocurrents is affected by the choice of material, but since the light spot position is determined from normalized photocurrents, it will not affect linearity and resolution results.

II. ANALYTICAL MODEL

For the steady-state and fully reverse-biased p-n junction, Woltring [12] showed that the following equation describes the potential distribution in the planar x - y resistive layer:

$$U_{xx}(x, y) + U_{yy}(x, y) = -\frac{\rho}{w} I_s(x, y) \quad (1)$$

where $U(x, y)$ is the electric potential, ρ the resistivity, w the thickness, and $I_s(x, y)$ the generated photocurrent density. If a small light spot is projected onto the sensitive surface, the generated photocurrent density may be approximated by the Dirac function:

$$I_s(x, y) = I_0 \delta(x - x_0) \delta(y - y_0). \quad (2)$$

For an isolated part of the boundary we may assume an infinite terminating impedance so Neumann-type boundary conditions should be applied (i.e., $\partial U / \partial n = 0$, where n is the normal to the boundary). At points of electrical contact we assume zero-terminating impedance and apply the Dirichlet-type boundary condition, $U = 0$.

After the potential distribution is determined by solving (1) with boundary conditions appropriate to each PSD geometry, the photocurrents at each electrode may be found from

$$I_i = \int \frac{\partial U}{\partial n} dL_i \quad (3)$$

where the integration is carried out over the length of each electrode, L_i . It is these photocurrents which, when combined appropriately, yield the normalized x and y position signals of the light spot. For example, in the duolateral PSD, the x position of the light spot is obtained by taking the difference of the photocurrents to the two electrodes

normal to x , and normalizing by the sum of the two photocurrents.

In what follows, output signals are numerically obtained for various PSD types for x -direction lines traversed by a light spot at evenly spaced y positions. Ideally the output plot should be equidistant straight lines parallel to the x axis. For the duolateral PSD Woltring's analysis predicts that this ideal is obtained [12], and, therefore, it is not necessary to simulate the performance. For the tetralateral and pincushion geometries the numerical method used was a canned program with the ability to solve two-dimensional linear elliptic partial differential equations in a nonrectangular domain with general boundary conditions. For the clover geometry, the equation solver could not be used and a boundary element method discussed in Brebbia [15] was used instead. Comparisons of the results from the different solution methods show good agreement.

For each PSD geometry considered an integration routine was used to determine the photocurrents to the contacts from the photopotentials. Then these photocurrents were combined in a manner appropriate to each different PSD type to determine the normalized x output signals as a function of the normalized actual light spot x location. In order to numerically characterize the deviation of the PSD from perfect linearity, a simple definition of distortion proposed by Petersson and Lindholm [16] was used:

$$D = \frac{S(y) - T(x, y)}{S(y)} \quad (4)$$

where $T(x, y)$ is the normalized y position signal predicted for the device, $S(y) = ky$ for all x is the idealized normalized linear output signal, and

$$k = S_y(y) = T_y(0, 0) \quad (5)$$

where k is defined as the inherent sensitivity of the PSD, and the origin is the PSD center. Note that the inherent sensitivity has units of inverse length and is not the same as the PSD resolution. The resolution depends on the amplitude of the photocurrents relative to the noise as well as the inherent sensitivity.

III. TETRALATERAL GEOMETRIES

Four distinct PSD electrode geometries were examined: the tetralateral, a modified tetralateral, the pincushion, and a modified pincushion we refer to as a clover. Since the tetralateral PSD photocurrents can be determined analytically [12], the numerical simulation was compared to the closed-form solution to verify that the equation solver was working. In this simulation we neglected the effect of the small gap between adjacent contacts. The closed form and numerical results agreed well. From the photocurrents to each electrode, the x and y normalized position signals (assuming unity gain) were found from the following:

$$x = \frac{I_{x=L/2} - I_{x=-L/2}}{I_{x=-L/2} + I_{x=L/2}} \quad (6)$$

$$y = \frac{I_{y=L/2} - I_{y=-L/2}}{I_{y=-L/2} + I_{y=L/2}} \quad (7)$$

where the subscripts indicate the locations of the electrodes. Shown in Fig. 4 is the normalized x signal output as a function of the normalized actual light spot x location for 11 equidistant x -direction lines with $L/12$ spacing on a conventional L by L tetralateral PSD. For this and subsequent figures the device effective area is assumed to be a central square of side length $10L/12$. From Fig. 4 we can see that the PSD is only linear over a very small region of the active area; the distortion becomes serious toward the boundary of the sensitive surface. The inherent sensitivity and maximum distortion are given in Table I, as well as the fraction of the effective area that has distortion below a set amount. Also given in Table I are the distortion and inherent sensitivity figures for the duolateral pincushion, and clover PSD's. Note that the duolateral PSD is theoretically distortionless and has an inherent sensitivity corresponding to a normalized output that varies from 0 to 1 as a light spot traverses an x or y line from the center to an edge.

As an intermediate step between the tetralateral PSD and the pincushion PSD, let us modify the conventional tetralateral PSD by bending the electrodes around the corners of the square sensitive area as shown in Fig. 5, while keeping the area covered by the electrodes the same as in the conventional tetralateral. The photopotential distribution is unmodified but the photocurrents to the electrodes are changed. From these photocurrents the normalized x and y outputs are found from the following equations [13]:

$$x = \frac{(I_1 + I_2) - (I_3 + I_4)}{I_1 + I_2 + I_3 + I_4} \quad (8)$$

$$y = \frac{(I_2 + I_3) - (I_1 + I_4)}{I_1 + I_2 + I_3 + I_4} \quad (9)$$

where I_i is the photocurrent to electrode i (labeled in Fig. 5).

Fig. 6 shows the x signal output as a function of the x spot location for the modified tetralateral PSD surface. Note that the very simple change in electrode geometry has caused a marked change in the PSD response. This modified tetralateral PSD is more linear than its conventional counterpart in the central part of the PSD, but the linearity decreases toward the periphery. In the central part the inherent sensitivity is given by $k = 2.4/L$, and the maximum distortion is given by $D_{\max} = 0.15$. In the outer part, $k = 6/L$ and $D_{\max} = 0.64$, substantially worse than the conventional tetralateral.

In the modified tetralateral PSD the generated photocurrent is divided into only two parts to get the x and y signals, meaning that if the power of the light source and the quantum efficiency are same as in the conventional tetralateral PSD, the output position signal is doubled in magnitude. Because the resolution of the sensor is always limited by the signal-to-noise ratio of the sensor cell, this

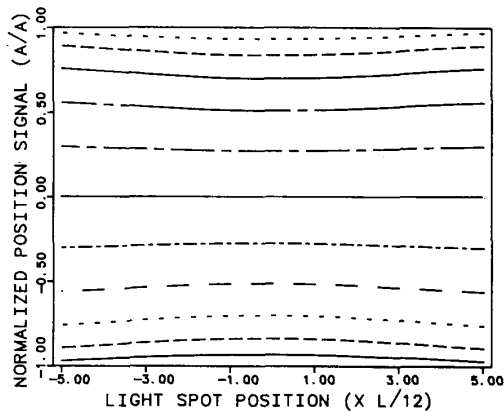


Fig. 4. Response of an L by L tetralateral PSD: output normalized position versus actual position for eleven equidistant horizontal lines.

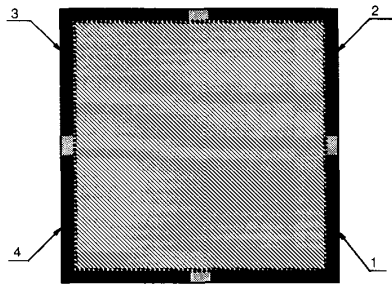


Fig. 5. Electrode geometry for a modified tetralateral PSD.

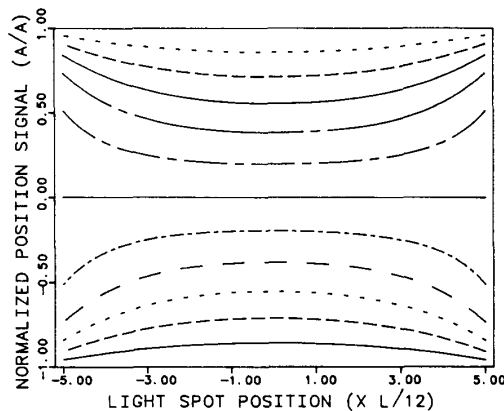


Fig. 6. Response of an L by L modified tetralateral PSD.

TABLE I
COMPARISON OF VARIOUS LATERAL EFFECT POSITION SENSITIVE DEVICES

Device	k ($\times 1/L$)	D_{max} (%)	Percent of area with distortion less than									
			1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
duolateral	2.0	0	100	100	100	100	100	100	100	100	100	100
tetralateral	3.3	33	6.0	10.4	13.1	16.2	18.8	21.7	24.6	27.5	30.3	33.4
pin cushion	0.2	8.4	6.9	13.5	20.8	31.6	42.8	53.6	66.6	86.1	100	100
clover	1.1	6.0	17.1	52.7	67.6	76.4	86.9	100	100	100	100	100

means that the resolution of the modified tetralateral PSD is double that of the conventional tetralateral PSD given a certain noise level.

The linearity of the modified tetralateral PSD can be

improved by reducing the lengths of the electrodes because the shorter these lengths, the smaller the amount of crosstalk, but there is a lower limit determined by the requirement for fully reverse-biased operation. If the electrode lengths were permitted to approach the lower limit condition of four point contacts at the corners, and if the fully reverse-biased condition still held, then the spaces between the lines in Fig. 6 would be about equal but the lines would be curved rather than straight. In order to produce equally spaced straight lines one might consider placing point contacts in the corners of a lateral effect PSD with curved boundaries. This produces the pincushion lateral effect PSD.

IV. PINCUSHION GEOMETRY

Let us consider the pincushion PSD. For proprietary reasons the exact shape of the boundaries of pincushion PSD's is not published in the open literature. We assume that each side of the planar PSD is bounded by identical circular curves with radius of curvature R as shown in Fig. 3. This assumption simplifies the numerical solution and appears to be a good approximation to the actual pincushion geometry.

For the curved pincushion boundaries, it is necessary to express the boundary conditions in terms of x and y coordinates. By defining the origin of the coordinate system to be at the geometric center of the PSD as shown in Fig. 3, we write the boundary conditions as

$$\frac{x+a}{R} U_x + \frac{y}{R} U_y = 0 \text{ for section } BC \quad (10a)$$

$$\frac{x}{R} U_x + \frac{y+a}{R} U_y = 0 \text{ for section } CD \quad (10b)$$

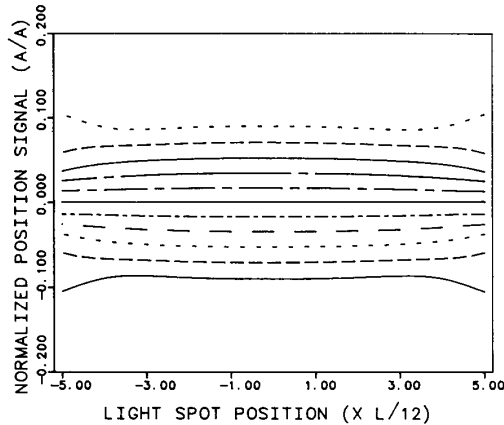
and

$$\frac{a-x}{R} U_x + \frac{y}{R} U_y = 0 \text{ for section } DA \quad (10c)$$

$$-\frac{x}{R} U_x + \frac{a-y}{R} U_y = 0 \text{ for section } AB \quad (10d)$$

where a is the distance from the origin of the PSD to the curvature center of the circular boundary, and points A , B , C , and D are labeled in Fig. 3.

In deriving the performance of the pincushion PSD, a fully reverse-biased condition has been assumed for the pincushion PSD although it may be difficult to achieve in practice. (See discussion of Wallmark device in Woltring [12].) After the photocurrents from each electrode were obtained, (8) and (9) were used to get the normalized x and y position signals, respectively, where the electrodes are numbered in Fig. 3. Shown in Fig. 7 is the calculated response of a pincushion PSD. Compared to Figs. 4 and 6, it is clear that the pincushion PSD is linear over a much broader range of its active surface than the tetralateral PSD's. However, it should be noted that the linearity dramatically decreases in the PSD corners. Table I shows that the pincushion is much more linear than the tetra-

Fig. 7. Response of an L by L pincushion PSD.

lateral PSD, but has a substantially lower inherent sensitivity.

V. THE CLOVER PSD—AN IMPROVED GEOMETRY

To improve the performance of the pincushion PSD, a modified pincushion design using eight protruding point contacts and four short contacts at the corners can be constructed. In this clover design the extended contacts at the corners are wired to point contacts halfway along each connecting boundary, as shown in Fig. 8.

For the x output signal in the clover PSD the contributions from the two neighboring point contacts of each contact pair along the primarily y -oriented boundaries will cancel because over most of the sensitive area, the resistances from the incident position of the light beam to each of the two neighboring point contacts are of almost same value. We have found numerically that if all these point contacts are on the curved boundaries, the assumption of cancellation is valid for a device with effective size of L by L as long as the light source is at least $L/12$ away from the boundary. If the point contact pairs are slightly offset from the boundaries, as shown in Fig. 8, then this assumption will be true for essentially all of the sensitive area.

The response of the clover design is shown in Fig. 9. This response was obtained by using (3) to determine the photocurrents to each contact, and (8) and (9) to provide the normalized x and y signals. As shown in Table I, the inherent sensitivity of the clover PSD is about five times that of the pincushion PSD while the maximum distortion is lower. Note that the conventional pincushion and clover designs are both nearly as linear as the duolateral PSD, but compared to the duolateral, the clover PSD has a faster response time, lower dark current, easier fabrication, and a bias that can be applied easier. For a given noise level, the resolution of the clover PSD is roughly twice that of the tetralateral PSD and about equal to that of the duolateral PSD. The only disadvantage when compared with the duolateral geometry is the reduced inherent sensitivity. It is likely that the linearity of the clover design may

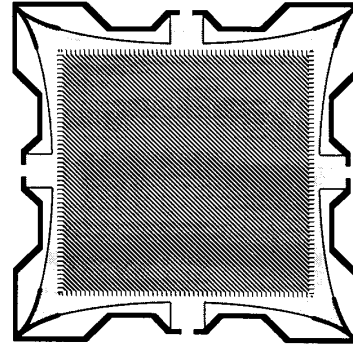
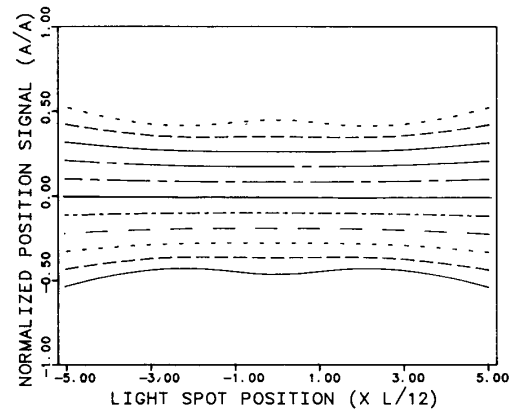
Fig. 8. Geometry of a clover PSD with an L by L effective area.

Fig. 9. Response of the clover PSD.

be further improved by optimum choice of size and locations along the boundaries of the point contact pairs, the radius of curvature of the boundary, and the lengths of the electrodes at the corners. The optimum design will not depend upon size, since the electrode structure connecting the protruding point contacts to the corner contacts will scale with the device effective area.

Theoretically the performance characteristics of the clover PSD are an improvement over those of any PSD currently in existence. This new geometry should combine the best features of the duolateral and tetralateral PSD's without their respective disadvantages.

VI. SUMMARY AND CONCLUSIONS

We have investigated the linearity, sensitivity, and resolution characteristics of a new type of lateral effect PSD (named the clover) and compared it to the duolateral, tetralateral, and pincushion PSD's. Of the conventional PSD geometries, the duolateral is the most linear and has the highest resolution, but is the slowest and has the highest dark current. The conventional tetralateral geometry has been shown to be linear only over a small central region of its sensitive surface, and have roughly half the resolution of the duolateral PSD's, although it has a sensitivity higher than that of the duolateral. The pincushion PSD is more linear than the tetralateral PSD although less

linear than the duolateral PSD. However, the pincushion geometry suffers from surface recombination induced signal loss due to difficulty in obtaining full reverse-biasing, and a low inherent sensitivity.

In light of the goal of achieving high linearity, high inherent sensitivity, high resolution, and fast response time, while maintaining an ability to operate in a fully reverse-biased condition, we have designed a new PSD which has a clover geometry. In this lateral effect PSD a pincushion geometry is modified to include an additional four pairs of point contacts located on or near the centers of the curved boundary lines. The contacts at corners are each connected to the two nearest point contacts along the boundary. We have shown that the clover PSD is nearly as linear as the duolateral PSD while holding similar position resolution, but has all of the contacts on a single side of the p-n junction so that the dark current is lower, response time is faster, reverse-bias can be applied easier, and fabrication cost is lower. The clover inherent sensitivity is five times that of the pincushion, and more than half that of the duolateral duoaxis PSD.

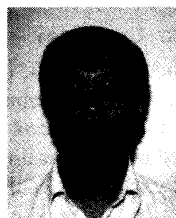
ACKNOWLEDGMENT

The authors wish to thank M. Jeong, R. Pelrine, W. Keene, and R. Katkhouda for their helpful comments.

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